

# Artificial Intelligence Agents: Learning to Reason and Act in Multimodal and Embodied Environments

图文讲义版：用原 PPT 作为视觉锚点，按“概念故事线 + 直觉解释 + 考试速记”整理。

先接上 L28：经典 agent 仍是 sensors 感知环境、actuators 改变环境；这节课把 sensor 从文字扩展到 vision+text → 为什么要 VLM：现实任务包含截图、图像、视频、表格、GUI；直接看视觉输入可以避免把整个世界硬塞进文本上下文 → 落地对象 CUA：Computer Use Agent 根据语言指令控制 OS 应用，执行点击、输入、滚动、终止等结构化动作 → 形式化核心：CUA 决策是一条 trajectory，模型反复观察状态  $s_t$ 、预测动作  $a_t$ ，直到 goal state 和 termination action → 数据从哪里来：AgentNet Tool 记录人类操作电脑的屏幕、鼠标键盘和 accessibility tree，再人工验收任务质量 → 为什么要 curate：原始视频太密、太噪，必须 action reduction 和 state-action matching，抽成能监督下一步动作的关键状态 → 为什么 naive finetune 不够：只喂  $I$ 、 $s_t$ 、 $a_t$  会缺 grounding 和 trajectory reasoning，因此 OpenCUA 加 long-form CoT → 训练 recipe：L1/L2/L3 输出格式、action history、visual history、QwenVL-2.5 规模、Stage 1 grounding 与 Stage 2 planning/reasoning → 评测与未来：OSWorld-Verified、Online-Mind2Web、ComputerUseArena 暴露真实环境变化、长程任务、循环、backtracking、安全和鲁棒性难题

## Benefits of a VLM-based AI Agent?

- In the previous lectures, we assumed that our agents would make decisions purely based on textual inputs
- Our world is inherently **multimodal**: our agents need to process much more than just textual inputs
  - Images
  - Videos
  - Graphs
  - Tables
- By leveraging a VLM, we can:
  1. Have access to pretrained knowledge available on the Web
  2. Use text/code to represent outputs
  3. Use vision directly avoiding context saturation; Why?

## 这节课一句话

这节课一句话：L29 把 L28 的 text-only agent 推出聊天框，变成能直接“看屏幕、想步骤、点按钮、打字、停止”的 Computer Use Agent；核心不是给 LLM 多写几句 prompt，而是用 VLM 读取视觉状态，把任务建模成 instruction-state-action trajectory，再用 AgentNet/OpenCUA 的数据整理、long-form CoT、GUI grounding、OS planning 和严苛评测，把一个会看图的模型训练成能在真实电脑环境里连续行动的系统。

## 1. 从 text-only 到 VLM: agent 终于长出眼睛

### Benefits of a VLM-based AI Agent?

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原 PPT 第 7 页

## 人话直觉

L28 的 agent 很像一个只听口述的助理：你把世界描述给它，它再用文字计划和调用工具。L29 往前迈了一步：让助理直接看桌面、看网页、看图表、看按钮。直觉上，这就像不要再让朋友通过电话给你念整个网页布局，而是直接把截图发给你；很多信息一眼就懂，硬翻成文字反而又长又容易漏。

**精确定义 / 机制：** Slide 7 的关键是把 agent 的输入空间从 pure text 扩展到 vision+text。现实世界和电脑环境天然包含 images、videos、graphs、tables、GUI layouts 等非文本状态；如果继续把所有状态转成文本，既会丢失空间布局、颜色、图标、按钮位置等 grounding 信息，也会造成 context saturation。VLM-based AI agent 的输入可以是视觉加文本，输出仍可用 text、code 或结构化 action 表示，因此保留了语言作为 planning/action interface 的优势，同时让 perception 更贴近真实环境。这里不要误解为“VLM 自动解决所有问题”：VLM 提供视觉感知能力，但 agent 仍需要动作空间、历史管理、训练数据和评测机制。

**考试问法：**如果考“why VLM-based agents are needed”，先写 text-only agents rely on textual inputs；再写现实任务包含图像、视频、图表、表格和 GUI；最后写 VLM 的三点收益：利用 web-scale pretrained knowledge、仍能用 text/code/action 表示输出、直接处理视觉输入以减轻 context saturation。

## 2. CUA 定义：不是帮你搜，而是替你操作电脑

### Computer Use Agent (CUA)

- CUA are agentic solutions that **can control your computer** by interacting with all the applications available
- Given a language command, they **can derive a plan** to execute all the actions required to complete the task successfully
- This goes beyond searching for documents given a search query
  - By giving agents access to the OS applications, we can enable agents to perform complex workflows required for real world applications
  - It facilitates the execution on tasks on a variety of different platforms as well (e.g., different operating systems)
- In the limit, we would like to have fully autonomous agents that can **independently execute tasks on our behalf!**

原 PPT 第 10 页

#### 人话直觉

Computer Use Agent 可以想成一个坐在你电脑前的实习生。你不是问它“蓝色牛仔裤有哪些”，而是说“去 Amazon.co.uk 找一条合适的蓝色牛仔裤”；它要打开网站、看页面、点筛选、输入关键词、比较结果，最后知道任务完成。搜索 agent 只给你信息，CUA 要在应用里把流程真的走完。

**精确定义 / 机制：**CUA 是能够控制电脑并与应用程序交互的 agentic solution。给定 language command，它需要 derive a plan，并执行一串必要动作来成功完成任务。这比普通搜索复杂，因为它暴露在真实 OS 和应用生态中：不同网站、文件管理器、浏览器、办公软件、系统设置都有不同界面和交互规则。CUA 的长期目标是自主代表用户执行真实 workflow，例如跨平台查找、填写、下载、整理、提交或配置。它的价值来自让 agent 接入 OS applications，但风险也随之上升：动作不再只是生成一句话，而是可能真实点击、输入、删除、购买或泄露信息。

**考试问法：**如果考“what is a Computer Use Agent”，一句话答：CUA is a vision-enabled agent that observes the computer screen and controls OS applications through structured actions to complete a language-specified task. 高分补充：它超越 search/query，因为要跨应用、多步骤、连续决策，并最终希望 autonomous execution on behalf of users。

### 3. Trajectory 建模：CUA 的每一步都在“看屏幕 → 出动作”

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#### The agent decision-making process

- We model the decision-making process of an agent as a **trajectory**:

$$(\mathcal{I}, \langle s_0, a_0 \rangle, \langle s_1, a_1 \rangle, \dots, \langle s_T, a_T \rangle)$$

- Where  $\mathcal{I}$  is a language instruction or command for the agent (e.g., “find a pair of blue jeans on Amazon.co.uk”) and  $s_0$  is the initial state of the agent
- The agent sequentially predicts an action  $a_i$  after *perceiving* the state of the environment  $s_i$
- It does so until it reaches the goal state  $s_T$  and performs a termination action  $a_T$
- **Vision-based agents**: We assume that the *state* is a screenshot of the *screen* that the agent can see.
  - No DOM or HTML of the page!

Wang, Xinyuan, et al. "OpenCUA: Open Foundations for Computer-Use Agents." NeurIPS 2025.

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#### 人话直觉

把 CUA 想成玩密室逃脱：任务说明是“找到蓝色牛仔裤”，当前房间状态是屏幕截图，你每次只能做一个动作——点哪里、输入什么、滚动多少、是否结束。做完动作后房间变了，你再看新屏幕，再决定下一步。会不会成功，取决于整条路径，而不是某一步回答得漂不漂亮。

**精确定义 / 机制**：Slide 14 把 agent decision-making process 建模为 trajectory：语言指令  $\mathcal{I}$  后面跟着状态和动作交替出现。 $s_t$  是第  $t$  步环境状态；对 vision-based CUA 来说，状态通常是屏幕截图，而不是网页 DOM 或 HTML。 $a_t$  是模型在观察当前状态和任务目标后预测的动作，例如 click、type、scroll、press key 或 terminate。这个设定的重要性在于，它把 CUA 从“单轮图像问答”变成 sequential decision-making：模型必须理解当前截图、任务目标、历史动作和未来目标状态之间的关系。Termination action 也很关键，因为不会停的 agent 会循环、重复点击或做多余操作。

$$\mathcal{I}, s_0, a_0, s_1, a_1, \dots, s_T, a_T$$

**考试问法**：如果考“formalise CUA decision process”，写出 trajectory，并解释  $s_t$ 、 $a_t$ 、goal state、termination action。一定要强调 vision-based setting 中 state is screenshot, not DOM/HTML；这让任务更真实，但也更依赖 visual grounding。

## 4. AgentNet 数据整理：从人类录像里炼出可学习的动作监督

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### How do we curate the data?

- Remember that AgentNet Tool collects directly screen recordings which have high-frequency frame rates associated with real humans completing tasks
  - Can our agents learn from this video data? No!
- Therefore, the authors created a careful data processing pipeline that ensures
  - Action reduction**: reduce the density of actions and focus on high-level signals only (e.g., for “scroll” actions we record only the start and end position)
  - State-action matching**: extract key video frames that are useful to predict the action at the current timestep
- After the final action, they append a final termination frame that determines the last state of the agent

Wang, Xinyuan, et al. "OpenCUA: Open Foundations for Computer-Use Agents." NeurIPS 2025.

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### 人话直觉

如果你把一个人操作电脑的完整录像直接丢给模型，就像让学生看 60fps 的做菜视频，却不告诉他哪一帧对应“加盐”、哪一段只是手在移动。模型会被鼠标抖动、连续滚动、过渡帧淹没。训练 CUA 要做的是把录像剪成教材：这一屏重要，下一步动作是这个，完成后停在那个状态。

**精确定义 / 机制：** AgentNet Tool 收集的是人类 annotators 在电脑上完成任务的交互数据，包括 screen recording、mouse/keyboard tracking 和 accessibility trees；但 slide 17 强调，原始高频 screen recordings 不能直接作为高质量训练样本。OpenCUA 需要一个 curation pipeline。第一步 action reduction：降低动作密度，只保留高层有用信号，例如 scroll 不记录每个微小滚轮事件，而记录起点和终点。第二步 state-action matching：从视频中抽取能够预测当前动作的关键帧，并与对应动作配对。最后还要 append final termination frame，让模型看到“任务完成时的最终状态”，学习什么时候停止。

**考试问法：**如果考“why raw AgentNet videos cannot directly train CUA”，答视频帧率高、动作密、低层噪声多；然后写两个 pipeline 关键词：action reduction 和 state-action matching；最后说明 final termination frame 帮助模型学习停止条件。

## 5. Long-form CoT: 不是让模型胡思乱想，而是补上观察和计划的中间监督

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### Long-form Chain-of-Thought for CUA

- To incentivise the model to reason about the task and what it sees on the screen, they synthesise a dataset that including long-form Chain-of-Thought data
- They structure their approach in three different levels:
  - Level 3: focus on salient textual and visual elements
  - Level 2: reflective reasoning focusing on planning next actions, reflecting on previous steps
  - Level 1: model predicts the action which is now grounded in the reasoning trace as well
- They design a synthetic generation pipeline by using Claude-3.7 Sonnet

Alessandro Suglia

Wang, Xinyuan, et al. "OpenCUA: Open Foundations for Computer-Use Agents." NeurIPS 2025.

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#### 人话直觉

只给模型“截图 → 动作”，像教新手开车时只说“现在打方向盘”，却不解释“我看见左侧有车、前面是路口、所以要减速再转”。OpenCUA 的 long-form CoT 更像教练旁白：先指出屏幕上哪些文字和按钮重要，再反思上一步做了什么，最后才给出下一步动作。

**精确定义 / 机制：** Slide 19 说明 naive finetuning 可以把  $(s_t, a_t)$  当作训练数据，但 OpenCUA 发现效果不好，原因是模型缺乏 grounding 和 trajectory reasoning。Slide 20 的 long-form Chain-of-Thought 解决思路是合成带中间推理的数据，让模型显式连接任务、屏幕元素、历史状态和下一步动作。它分三层：Level 3 关注 salient textual and visual elements，相当于 observation；Level 2 做 reflective reasoning，包括 planning next actions 和 reflecting on previous steps；Level 1 输出 action，此时动作被前面的 reasoning trace grounding。讲义还提到 synthetic generation pipeline 使用 Claude-3.7 Sonnet 生成这些 reasoning traces。考试里要把它理解成 training supervision，而不是简单的“考试时输出越长越好”。

**考试问法：** 如果考“why does OpenCUA use long-form CoT”，答 naive finetuning lacks grounding and trajectory reasoning；long-form CoT provides intermediate supervision over observation, planning/reflection and action. 可列 L3=salient visual/textual elements, L2=planning/reflection, L1=grounded action prediction。

## 6. OpenCUA training recipe: 先看准 GUI, 再学会在 OS 里做事

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### The OpenCUA Training Recipe

- They use a QwenVL-2.5 model as starting point for their experiments; They finetune three different model sizes:
  - 7B
  - 32B
  - 72B
- They explored different training regimes that use data during different phases
  - Stage 1 = visual grounding on GUI screens
  - Stage 2 = planning + reasoning in OS environment
- The authors report additional results for other ways of combining these two stages
  - It turns out that joint training is the best approach but more computationally intensive!

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Wang, Xinyuan, et al. "OpenCUA: Open Foundations for Computer-Use Agents." NeurIPS 2025.

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#### 人话直觉

训练 CUA 像训练一个新员工用电脑。第一阶段不是让他直接做复杂报销，而是先认清屏幕：按钮在哪里、输入框在哪里、菜单是什么意思。第二阶段才教他按任务目标规划步骤、记住刚才点过什么、从错误里调整路线。OpenCUA 的 recipe 就是在这两类能力之间配料。

**精确定义 / 机制：** OpenCUA 的 training recipe 不只是把 AgentNet trajectory 喂给 VLM。Slide 22 说明它设计了不同 reasoning trace formats: L1 是 action only, L2 是 thought + action, L3 是 observation + thought + action; 还加入 action history 和 visual history, 例如 previous actions 和 previous three screenshots, 以帮助模型理解已经发生过什么。Slide 24 进一步说明实验从 QwenVL-2.5 出发, finetune 7B、32B、72B 等规模, 并比较不同训练 regime。Stage 1 是 GUI screen visual grounding, 重点是看懂屏幕元素; Stage 2 是 OS environment 中的 planning + reasoning, 重点是多步任务执行。Joint training 效果最好但计算更贵; Stage 2 only 对低计算场景更现实, 尤其当现代 VLM 已经具备一定 GUI grounding 能力时。

**考试问法：** 如果考“describe OpenCUA training pipeline/recipe”, 按 data → CoT format → history inputs → training stages 答。关键词必须有 L1/L2/L3、action history、visual history、QwenVL-2.5、Stage 1 GUI grounding、Stage 2 planning + reasoning, 以及 accuracy/inference speed 或 compute trade-off。



## 7. OSWorld-Verified: 真实电脑评测会被世界变化反噬

### OSWorld-Verified

- Since the initial release in 2024, OSWorld has become an industry standard for CUA evaluation with many giants including OpenAI and Anthropic using it as a benchmark environment for their frontier models
- However, people have identified important weaknesses of the original data collection that have had an effect on model performance
- **OSWorld-Verified** is the current instantiation of the benchmark which includes:
  1. **Infrastructure**: Migrated to AWS cloud for massive parallelization (10+ hours → minutes)
  2. **Task Quality**: Fixed 300+ issues including web structure changes, instruction ambiguity, and evaluation robustness
  3. **Evaluation**: Established public evaluation platform for verified apple-to-apple comparisons
  4. **Performance**: Current leading models show success primarily stems from extensive human trajectory data

Xie, Tianbao, et al. "OSWorld: Benchmarking multimodal agents for open-ended tasks in real computer environments." NeurIPS 2024.

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原 PPT 第 29 页

#### 人话直觉

评测 CUA 不像评测静态选择题。网页会改版，CSS class 会变，机票日期会过期，按钮文案会换，模糊指令可能有多个合理答案。一个 benchmark 如果不维护，就像用去年商场地图考今天的导航员：模型可能没错，是地图过期了。

**精确定义 / 机制：** Slide 29 介绍 OSWorld-Verified: OSWorld 已成为 CUA evaluation 的重要 benchmark，但原始数据存在会影响模型表现的问题，因此需要 verified instantiation。改进包括 infrastructure 迁移到 AWS cloud，使并行评测从十多个小时缩短到分钟级；task quality 修复 300+ issues，包括 web structure changes、instruction ambiguity 和 evaluation robustness；建立 public evaluation platform 以支持 apple-to-apple comparison；并观察到当前领先模型的成功很大程度来自 extensive human trajectory data。Slide 30 进一步列出具体失效源：网站 HTML/CSS/content 变化、time-sensitive booking tasks 过期、指令歧义、evaluation function 过严或不鲁棒。

**考试问法：**如果考“why OSWorld-Verified matters”，答它修复原 OSWorld 中环境变化、任务歧义和评估脆弱问题，使 CUA 评测更可靠、更可复现、更公平。高分补充：CUA benchmark 的难点来自真实世界不是静态数据集。



## 8. CUA evaluation: success rate 只是总分，不能解释为什么翻车

### How do we evaluate CUA?

- Evaluation of CUA agents requires assessing whether the agent can reach the final goal given a language instruction
- Success rate metrics are useful to determine how often agents can successfully complete the tasks
- It is also important to use specialised datasets to assess the ability to perform **visual grounding** (e.g., select a given button!)
- However, CUA tasks typically involve a very high number of steps which complicates the evaluation setup. Many important aspects to think about:
  - How do we reward intermediate goals?
  - How do we recognise when the agent is stuck in a loop?
  - How do we incentivise the agent's ability to backtrack?
  - If the agent fails, is it because of its inability to execute actions or due to its inability to locate specific elements of the desktop?

原 PPT 第 35 页

### 人话直觉

如果一个 agent 没完成任务，只看失败并不够。它可能第一步就没看见按钮，也可能中途点错后不会回退，也可能一直在同一页循环，还可能动作执行器本身点偏了。就像看学生解题，最后答案错了，你还要知道是概念错、计算错、审题错，还是中间抄错。

**精确定义 / 机制：**Slide 35 把 CUA evaluation 的核心问题说得很清楚：评估 agent 是否能根据 language instruction reach the final goal。Success rate 有用，因为它衡量任务成功比例；但 CUA tasks 通常步骤很多，最终成功率不足以诊断系统能力。评测还需要 specialised datasets 检查 visual grounding，例如能否选择指定按钮或定位文本框。长程任务还会带来中间目标奖励、循环检测、backtracking 激励和 failure attribution 等问题：agent 失败到底是因为不会执行动作，还是因为无法 locate specific elements of the desktop？这直接影响模型改进方向。

**考试问法：**如果考“why evaluating CUA is difficult”，答多步、长程、真实环境变化大；success rate useful but insufficient；还要评估 visual grounding、intermediate goals、loop detection、backtracking 和 failure cause attribution。

## 9. 未来挑战：CUA 真正难在可靠地长期操作真实电脑

### Future work on CUA

- An incredible opportunity space for researchers and practitioners to redefine the way we interact with computers and define computer-based workflows
- However, for production-ready systems we need to ensure reliability across a variety of axes:
  1. The visual grounding gap
  2. Long-horizon modelling
  3. Credit-assignment problem
  4. Security & Reliability

原 PPT 第 39 页

### 人话直觉

CUA 的酷点也是危险点：它真的能碰你的电脑。聊天机器人说错一句话还可以忽略，CUA 点错一个按钮可能删文件、泄露数据、买错东西或进入无限循环。未来研究不是只追求“更聪明”，而是追求“看得准、走得远、错了能追责、操作足够安全”。

**精确定义 / 机制：**Slide 39 总结 production-ready CUA 的四个轴。Visual grounding gap 指模型难以稳定理解屏幕元素、图标、按钮、可点击区域和文字之间的关系；看错位置会直接导致错动作。Long-horizon modelling 指多步任务中历史很长，错误会累积，模型需要记住目标、已完成步骤和当前状态。Credit-assignment problem 指最终失败后很难判断是哪一步 observation、plan 或 action 导致失败，这让训练和调试都困难。Security & reliability 在 CUA 中格外重要，因为 agent 具有真实 action capability，可能遭遇环境干扰、恶意网页、prompt injection、权限滥用或高风险操作。Slide 45 的 summary 把全课收束为：视觉让 agent 进入电脑和机器人等 embodied domains，而 language 成为 high-level planning 与 low-level action execution 的桥梁。

**考试问法：**如果考“future challenges for CUA”，按四点写：visual grounding gap、long-horizon modelling、credit assignment、security & reliability。每点至少解释一句，并强调 CUA 因为能真实操作电脑，所以安全和可靠性比普通文本 agent 更高 stakes。

### 考试速记版

- L29 主线：从 text-only agents 扩展到 multimodal/embodied agents，重点是 Computer Use Agents。
- VLM-based agent 输入可包含 vision+text，输出仍可用 text/code/structured actions 表示。
- 需要 VLM 的原因：现实环境有图像、视频、表格、图表和 GUI；直接看视觉输入能减少 context saturation。
- CUA 定义：根据语言指令观察屏幕并控制 OS 应用，执行点击、输入、滚动、终止等动作完成任务。

- CUA trajectory: instruction I 后接状态和动作交替序列; vision-based state 通常是 screenshot, 不是 DOM/HTML。
- Termination action 很重要: agent 必须知道任务何时完成, 否则会循环或执行多余动作。
- AgentNet Tool 收集 screen recording、mouse/keyboard input tracking、accessibility trees, 并覆盖 Windows、macOS、Ubuntu。
- 原始录像不能直接训练: 帧率高、动作密、噪声大; 需要 action reduction 和 state-action matching。
- AgentNet 规模: 22,625 条 human-annotated trajectories, 约 12K Windows、5K macOS、5K Ubuntu, 平均 18.6 steps。
- Naive finetune 用 (I, s\_t, a\_t) 效果不好, 因为缺 grounding 和 trajectory reasoning。
- OpenCUA long-form CoT: L3 observation/salient elements, L2 planning/reflection, L1 action。
- Training recipe: L1/L2/L3 数据混合, action history, visual history, QwenVL-2.5 作为 base model。
- Stage 1 = GUI screen visual grounding; Stage 2 = OS environment planning + reasoning; joint training 更强但更贵。
- CUA evaluation 不只看 success rate, 还要看 visual grounding、intermediate goals、loop detection、backtracking 和 failure attribution。
- OSWorld-Verified 的意义: 修复网站变化、指令歧义、评估函数脆弱等问题, 让评测更公平可复现。
- Open-ended evaluation 重要: ComputerUseArena 发现静态 benchmark 排名可能和真实开放任务表现不一致, 说明鲁棒性不足。
- 未来挑战四件套: visual grounding gap、long-horizon modelling、credit assignment、security & reliability。

### 一段稳妥考场回答

Computer Use Agents are vision-enabled AI agents that can operate a computer environment on behalf of a user. Unlike text-only LLM agents, they do not rely only on textual inputs: a VLM can observe screenshots or other visual states while language remains the bridge between high-level instructions and low-level structured actions. A CUA is commonly modelled as a trajectory consisting of a language instruction, states and actions, for example an instruction followed by  $s_0$ ,  $a_0$ ,  $s_1$ ,  $a_1$  and so on until a goal state and a termination action. In the vision-based setting, each state is usually a screenshot rather than DOM or HTML, and actions can be represented as structured functions such as click, type, scroll, key press or terminate. OpenCUA shows how such agents can be trained from human computer-use trajectories. AgentNet collects screen recordings, mouse and keyboard traces and accessibility trees across Windows, macOS and Ubuntu, but raw videos are too dense and noisy to train from directly. Therefore the data must be curated using action reduction, which keeps high-level useful actions, and state-action matching, which pairs key frames with the actions they support; a final termination frame teaches the agent when to stop. Simply finetuning a VLM on instruction, state and action triples is not enough because the model lacks visual grounding and trajectory reasoning. OpenCUA therefore adds long-form Chain-of-Thought supervision: Level 3 describes salient visual and textual elements, Level 2 performs planning and reflection over previous steps, and Level 1 predicts the grounded action. Its training recipe also uses action history, visual history, GUI screen grounding and OS-level planning/reasoning stages, with trade-offs between accuracy, inference speed and compute. Evaluating CUA is difficult because long multi-step tasks can fail for many reasons: poor visual grounding, loops, inability to backtrack, wrong intermediate actions, brittle environments or weak evaluation functions. Benchmarks such as OSWorld-Verified improve reliability by fixing ambiguous or broken tasks, while open-ended settings such as ComputerUseArena reveal robustness problems that static benchmarks may hide. The main future challenges are closing the visual grounding gap, modelling long-horizon tasks, solving credit assignment when failures occur, and ensuring security and reliability because CUA agents can perform real actions on real computers.